

BCS Practitioner Certificate in Requirements Engineering

Classroom Activities: Trainer Answer Key

Activity 1: Spot the Problem (Topic 1)

Answer Key

R1. *The system shall be fast.*

Problem: Ambiguous and not testable. Fast means what? There is no measure and no criteria, so no tester could ever say whether it has been met.

Proper version (for later): The system shall return vehicle search results within 2 seconds.

Signpost: How to write measurable, testable requirements is covered in the Documentation topic.

R2. *The system shall allow the Sales Person to record a rental and record a return and produce monthly revenue reports and send marketing emails to previous customers.*

Problem: Multiple requirements bolted into one with "and". Four separate needs that cannot be individually prioritised, traced, tested or signed off. If one part fails testing, the whole requirement fails.

Proper version (for later): Split into four separately numbered requirements.

Signpost: One requirement, one need. Covered under documentation standards.

R3. *The system might allow customers to extend their rental if the Workshop Supervisor thinks it is a good idea.*

Problem: Vague, non-committal language. "Might" is not a requirement, it is a maybe. And "if the Workshop Supervisor thinks it is a good idea" is an opinion, not a rule. Nobody could build or test this.

Proper version (for later): The system shall allow rental extension subject to vehicle availability.

Signpost: Requirements use "shall". Note this also avoids confusion with the MoSCoW words Must and Should, which are priority labels, not requirement wording.

R4. *The system shall use a MySQL database with a three-tier architecture to store customer records.*

Problem: A solution dressed up as a requirement. MySQL and three-tier architecture are design decisions. The actual need is simply that customer records are stored. The requirement has taken a decision that belongs to the designers.

Proper version (for later): The system shall store customer records.

Signpost: Requirements state the what, design chooses the how. Expect the strongest debate here with a technical audience. Let it run for a minute, then land the line: "Nothing wrong with MySQL. The problem is the requirement has made the decision for the designers."

R5. *The system shall display available vehicles quickly and the search should be user-friendly.*

Problem: Two problems in one line. It is a compound requirement (display and search) and it is ambiguous twice over: "quickly" and "user-friendly" are both unmeasurable. Good groups will spot both, credit them for it.

Proper version (for later): Split into separate requirements, each with measurable criteria, for example a maximum response time and defined usability criteria.

R6. *The system shall allow the Sales Person to record a vehicle inspection.*

R7. *Only the Workshop Supervisor shall be able to record a vehicle inspection.*

Problem: R6 and R7 conflict. One says the Sales Person records inspections, the other says only the Workshop Supervisor can. Both cannot be true. This is the kind of contradiction that hides in a long requirements list and only surfaces when someone examines the set as a whole.

Resolution (for later): The conflict must be resolved with the stakeholders, most likely confirming inspection belongs to the Workshop Supervisor role.

Signpost: Finding and resolving conflicts is an analysis activity. "Resolving this is exactly what we cover in the Analysis topic tomorrow."

R8. *The system shall process refunds in accordance with the refund policy.*

Problem: It references a document that does not exist. There is no refund policy yet. You cannot build or test compliance with rules nobody has written down. It also has no clear source: who asked for this and who owns the policy?

Proper version (for later): Elicit the actual refund rules and state them in the requirement, or get the policy written and approved first, then reference it by name and version.

Signpost: Recording the source of every requirement is part of requirements management, which runs from day one.

Activity 2: Before, During, After

Answer Key

BEFORE: 1, 2, 3, 4, 5, 6. Agenda, venue, objective, technique choice, document-based question prep, participant selection. Items 3, 4 and 6 come straight off the workshop preparation list on Slide 29.

DURING: 7, 8, 9, 10, 11, 12, 13, 14. Introduction, scene setting, open question, probing question, staying on track, managing dominance, keeping a record, and the closing summary.

AFTER: 15, 16, 17, 18. Actions with owners and timescales, the write-up, the confirmation follow-up, and booking the return visit.

Activity 3: Choose Your Weapon – Technique Selection

Answer Key

Scenario 1: Protocol analysis (shadowing also strong). Tacit knowledge, skills. The matrix gives interviewing one star for skills, which the scenario has already proven, fifteen years of instinct cannot be talked out. Having him inspect a real van while narrating converts the skill into a checklist, which is literally the deliverable. Shadowing earns near-full credit, the edge for protocol analysis is the narration: you get his reasoning, not just his actions.

Scenario 2: Document analysis. Factor-driven: stakeholder availability is zero, so every people-based technique is eliminated regardless of stars. Document analysis needs nobody, harvests corporate explicit knowledge, and generates the question sets for next week. Bonus credit for any table saying the documents must later be verified against current practice.

Scenario 3: Workshop. The giveaways: a conflict to resolve between groups, and commitment to the outcome, consensus and buy-in, workshops' signature advantages. Separate interviews would collect two entrenched positions and leave the BA to arbitrate alone. Watch for a sharp table spotting this is the R6/R7 conflict from Activity 1 getting its resolution method, praise that loudly.

Scenario 4: Prototyping. Trigger in full view: customers cannot describe or envisage what they want, "just make it easy" is the blank page problem. Show a mock booking screen and vague wishes become specific reactions. The feature exists nowhere, so there is nothing to observe, killing shadowing. Scenario analysis is a reasonable runner-up and pairs well with the prototype.

Scenario 5: Observation (shadowing). Tacit, taken-for-granted, with a warning light in the wording: staff sincerely believe they follow the standard process, so asking them will faithfully return the official version. Only watching each branch reveals the real workarounds. Strong tables should add the caveats: being watched changes behaviour, and one visit shows one occasion, so observe more than once. If a table raises the travel implications of five branches, credit it, that is Slide 43 thinking.

Scenario 6: Interview. The deliberate twist, after five scenarios of "not interviews," this is interviews' home ground. The knowledge is explicit, she can state the rules precisely when asked, so interviewing's two stars apply, and the factors seal it: one senior stakeholder, thirty minutes, once, no repeat visits. A prepared, structured interview with questions ready in advance, documented immediately since there is no going back. This scenario exists to teach that the lesson was never "interviews are bad," it was "match the tool to the knowledge and the constraints."

Activity 4: The Sorting Office – Classify the Requirements

Answer Key

GENERAL: 2, 5, 8, 11. 2 is branding, THE trap, delegates will want to call it non-functional. 5 is a business constraint (budget, timescale, resources). 8 is legal, and note it is written as a pointer to the policy, the correct style for general requirements. 11 is business continuity, the business carrying on, not the system recovering.

TECHNICAL: 4, 7. Both contain the giveaway word "existing". 4 is a hardware constraint. 7 is an interface constraint, the mandated standard format. If challenged that 7 sounds functional, use the capability-or-compatibility test: the data flowing is a business need, but the mandated means and existing system make the constraint technical as written.

FUNCTIONAL: 1, 9. 1 is data entry, creating a new record. 9 is data maintenance, changing an existing one, and the pair tests exactly that distinction.

NON-FUNCTIONAL: 3, 6, 10, 12. 3 performance, 6 usability, 10 accessibility, 12 archiving and retention. 6, 10 and 12 are the frequent misclassifications, expect at least one table to put usability or accessibility under general.

Activity 5: From Conversation to Catalogue

REQ-021, Check vehicle availability online. Description: The system shall allow a customer to view the availability of vehicles for their chosen dates and vehicle category via the online booking pages. Type: Functional (retrieval). Priority: High. Acceptance criteria: a customer sees only vehicles genuinely available for the requested dates; availability shown matches the branch system's records.

REQ-022, Make an online booking. Description: The system shall allow a customer to book an available vehicle online for their chosen dates and receive a booking confirmation. Type: Functional (data entry). Priority: High. Acceptance criteria: a completed booking records customer, vehicle and dates; a confirmation is displayed and sent to the customer; the vehicle no longer shows as available for those dates.

REQ-023, Capture driving licence details. Description: The system shall capture the customer's driving licence details at the point of online booking. Type: Functional (data entry). Priority: High. Acceptance criteria: a booking cannot be completed without a licence number and expiry date; captured details are stored against the booking.

REQ-024, Licence verification compliance. Description: The rental process shall comply with driving licence verification legislation, as defined in AutoRent's licence checking policy. Type: General (legal), and note the correct pointer style, referencing the law and policy rather than restating them. Priority: High (legal obligations are rarely anything else). Acceptance criteria: every completed rental holds a verified licence record; compliance confirmed at audit.

REQ-025, Online response time. Description: The system shall display online page responses within 2 seconds for 95% of requests and within 5 seconds for 99% of requests. Type: Non-functional (performance). Priority: Medium or High, accept either with reasoning. Acceptance criteria: response time targets demonstrated under load testing.

REQ-026, Mobile usability. Description: The online booking pages shall be fully usable on smartphones; a first-time customer shall be able to complete a booking on a phone, unaided, within 5 minutes. Type: Non-functional (usability). Priority: High, Priya says most customers are on phones. Acceptance criteria: usability testing on common smartphone types confirms unaided completion within the target time.

REQ-027, Branch system integration. Description: The system shall integrate with the existing branch system using AutoRent's standard interface format, keeping bookings and availability synchronised. Type: Technical (interface, existing system). Priority: High, Priya was emphatic: no two separate systems. Acceptance criteria: an online booking appears in the branch system without manual re-entry; availability shown online matches the branch system.

REQ-028, Peak period capacity. Description: The system shall support peak summer Saturday demand across all five branches, up to 500 concurrent online users, without degradation of response times. Type: Non-functional (capacity and scalability). Priority: High. Acceptance criteria: under a simulated load of 500 concurrent users, 95% of page responses remain within the agreed response time target.

Model User Stories (required two, plus acceptable extras tables may produce)

- Required 1: As a customer, I want to check vehicle availability online, so that I do not have to phone the branch to find out.
- Required 2: As a customer, I want to book a vehicle online once I have found one available, so that I can secure it immediately without calling or visiting.
- Acceptable extra: As a customer, I want to enter my driving licence details during online booking, so that I can complete the legal checks upfront and collect the vehicle without delays at the branch.
- Acceptable extra: As a customer, I want the booking pages to work properly on my phone, so that I can book from wherever I am. If a table writes this, credit it and add the teaching point: this is really a non-functional requirement wearing story clothes; in Agile practice, qualities like mobile support usually live in the acceptance criteria of other stories rather than as stories of their own.

- If a table writes a staff-perspective story (for example: as a Sales Person, I want to see online bookings in my branch system, so that I can prepare vehicles), credit the correct format, then note the brief asked for the CUSTOMER perspective, precision in reading the brief is an exam skill too.

Marking guide: all three parts present, WHO is the customer (not "the user"), and the SO THAT expresses genuine value. Watch for circular whys like "so that I can book a vehicle" on the booking story, the why is the benefit: immediacy, no phone call.

Activity 6: Fix Them At Last

Answer Key

R1, "The system shall be fast" becomes: "The system shall return vehicle search results within 2 seconds for 95% of requests, and within 5 seconds for 99% of requests." Credit the percentage range especially, it is Slide 60's own fix pattern for the response-time example. An absolute single number ("within 2 seconds") is acceptable but the range is the better answer, and saying so reinforces the slide.

R5 must first be SPLIT, that is the trap, it is compound. It becomes two requirements: "The system shall display available vehicle search results within 2 seconds for 95% of searches" and "A first-time customer shall be able to complete a vehicle search unaided within 3 minutes." Marking essentials: the compound is split, both halves carry numbers, and "should" has become "shall". A table that fixes the wording but leaves it as one requirement has missed the trap, send them back to the Slide 57 definition: composite requirements must be separated.

Stretch, R3 becomes: "The system shall allow rental extension subject to vehicle availability." Shall form, condition stated, the Workshop Supervisor's opinion removed and replaced with a testable business rule.

Activity 7: Letter the Requirements (MoSCoW)

Answer Key (with the arguments, which are the marks)

1. **M.** Availability checks are the biggest driver of phone calls. Without this, the objective is dead on day one.
2. **M or S, the designed argument.** M case: unpaid online bookings generate a payment phone call, defeating the purpose. S case: pay-at-branch works short-term. Accept either WITH its reasoning, rule the debate with the discriminator: launched without it, does the objective fail?
3. **C.** Pleasant, sells nothing, saves no calls.
4. **M, the sleeper.** If staff cannot see online bookings, the channel collapses into MORE phone calls, Priya's "no two separate systems" rule. Watch for tables underrating it because it is invisible to customers.
5. **S.** Cancellation calls are common, so it matters, but launch survives briefly without it. Second increment.
6. **W, and the teaching moment.** Live chat staffed by branch teams moves phone workload to chat workload, it works AGAINST the objective. Won't have this time, revisit only if the objective changes.
7. **S, arguably M.** Without confirmation, anxious customers phone to check the booking worked. Accept M with that argument.
8. **W.** Future scheme, future delivery. Textbook Won't-have-this-time: acknowledged, recorded, deferred.

Activity 8: True or False – the Use Case Diagram

Answer Key

1. **FALSE.** No line connects Sales Person to Record Inspection, and the missing line IS the access rule, R7 enforced visually.
2. **TRUE.** Both actors have lines to Search Vehicles.
3. **FALSE.** Actors stand OUTSIDE the boundary. They interact with the system, they are not part of it. Classic exam distractor.
4. **FALSE, the most important one.** Use case diagrams show NO sequence, order or flow. Order needs a process model or activity diagram, Specimen Q9's exact lesson.
5. **TRUE.** Four ovals, and counting the ovals IS reading the scope.
6. **TRUE.** Only the Workshop Supervisor's line reaches that oval.
7. **FALSE.** No data appears on a use case diagram, data structure lives in the class model, which is precisely why the two are used together.
8. **TRUE.** The system boundary is the scope line: inside is the system's job, outside is not.
9. **FALSE.** No timing, performance, or anything non-functional appears on a use case diagram.
10. **TRUE.** Actors can be external systems, AutoRent's finance system could appear as an actor.

Activity 9: Spot the Classes

Model Answer – Two Acceptable Readings, Both Full Credit

Reading 1 (the simple reading, most tables will produce this): TWO classes.

- Customer: name, email, password
- Booking: startDate, endDate, agreedDailyRate

In this reading, the email and password are treated as facts about the customer, which is a perfectly fair reading of the scenario as written. Full credit.

Reading 2 (the finer reading, a sharper table may produce this): THREE classes.

- Customer: name
- Account: email, password
- Booking: startDate, endDate, agreedDailyRate

The reasoning behind the split: the scenario says the customer REGISTERS an account, so the account is arguably a thing in its own right that AutoRent holds data about. The login credentials, email and password, belong to the Account (they are facts about the account, not about the person: if the customer changed their account, their name would not change but their email and password might). The personal detail, name, stays on Customer. On a full model these two classes would be connected, Customer 1 to 0..1 Account, but connections are not part of this activity.

Activity 10: Interrogate the Class Model

Answer Key

1. **No.** Standing at Rental, the Customer end reads 1, mandatory. Anonymous rentals are impossible by design.

2. Every rental has at least one payment, the deposit at minimum, unpaid rentals cannot exist. 0..* would permit exactly that, so the minimum bound of 1 IS the business rule.
3. Zero while the vehicle is still out, the inspection has not happened yet; maximum one because each rental gets exactly one inspection at return.
4. **No.** Standing at Vehicle, the Branch end reads 1, exactly one branch per vehicle. Transfer means changing the branch, never having two.
5. **No.** 1..* requires at least one vehicle per category, an empty category is invalid. Change to 0..* IF the business wants pre-created categories, and the BUSINESS decides, not the analyst or developer. The model just generated an elicitation question, the generating-questions purpose of modelling firing live.
6. **No.** 0..* is unlimited. Fix: 0..3. Lesson: read every 0..* suspiciously, unlimited is itself a business decision, usually an accidental one.
7. **Payment (AdditionalDriver also creditable).** The use case diagram has Record Rental, Record Return, Record Inspection, Search Vehicles. Nothing takes a payment, yet the class model demands 1..* payments per rental. One model just exposed a hole in the other, a missing Take Payment use case. Doing this check systematically for Create, Read, Update, Delete is exactly what the CRUD matrix is, this question IS the bridge into the CRUD slide.

Activity 11: Who Is Speaking? (Topic 5, after Slide 87)

Answer Key

1. **Business owner.** Correctness for their area.
2. **Tester.** The testability check. Follow-up: what does that tell us about the requirement? It fails the quality checklist, validation just caught it.
3. **Business sponsor.** Objectives, scope and money. They chair the review and sign off.
4. **Solution architect.** High-level technical feasibility against the existing landscape.
5. **SME.** Specialist BUSINESS advice, legal and regulatory. Plant the Q31 flag here: SMEs give business advice, developers check technical feasibility. The exam swaps those words deliberately.
6. **Project office.** Standards, templates, version numbers. The paperwork check.
7. **Developer.** Detailed build complexity, technical feasibility, never business feasibility.
8. **THE TRICK, nobody legitimate.** The BA presents, answers questions and records issues. The BA does NOT approve or sign off, the business owns the need so the business owns the decision. If statement 8 happened in a real review, the chair should stop it.
9. **Business owner again.** And the bonus lesson: this is validation working perfectly, a small wording correction from the person who actually knows, caught before anything is built.

Activity 12: The Review Meeting (Topic 5, after Slide 91)

Answer Key

R-31: PASSES. Clear, testable, correct, no reviewer objects. Deliberately included so tables learn validation is not fault-finding theatre, good requirements pass.

R-32: FAILS, tester and business owner. "Appropriately" is untestable and ambiguous: the tester cannot write a test, and the business owner asks "appropriate according to whose rules?" Fix: state the refund rules or reference the approved policy. (Old friend R8 from day one, evolved.)

R-33: FAILS, SME. "Permanently" breaches data protection retention principles, the specialist business advice catch. Fix: a defined retention period then secure deletion.

R-34: FAILS, solution architect (developer creditable). Branch data syncs overnight in the current landscape; one-second real-time is not feasible as architected. Outcome: renegotiate with the requirement's owner (for example, availability accurate as at last sync, clearly timestamped) or accept an architecture change with its cost, and THAT is a sponsor-level decision, not the BA's.

R-35: PASSES. And a nice moment: it is the R6/R7 conflict, resolved, correctly worded, sailing through review. The course's longest-running requirement finally gets validated.

Activity 13: Formal or Informal, Agile or Linear? (Topic 5, after Slide 93)

The Six Situations and Answers

1. The completed AutoRent BRD is reviewed in a chaired meeting with the sponsor signing off. FORMAL LINEAR.

2. During backlog refinement, the team and Product Owner rework three user stories' acceptance criteria. INFORMAL AGILE, the ongoing, iterative validation.

3. Time pressure means the BA walks the sponsor through the key requirements at her desk and notes her comments. INFORMAL LINEAR.

4. At project initiation, stakeholders review and agree the initial solution scope and starting backlog. FORMAL AGILE.

5. A sprint review demonstrates the booking screen and the Product Owner accepts the stories against their acceptance criteria. INFORMAL AGILE (accept formal Agile with argument, sprint reviews have ceremony, the reasoning matters more than the pigeonhole).

6. The reworked BRD returns for its second chaired review after major changes. FORMAL LINEAR, and the callback: that is post-review outcome number one from Activity 12 playing out.

Activity 14: Name the Element (Topic 6, after Slide 97)

Answer Key

1. **Requirements identification.** The unique ID at the moment of capture.

2. **Requirements cross-referencing.** Recorded links between requirements.

3. **Origin (and 7 is ownership).** The pair from the same element, origin and ownership. Credit tables that give the element name once for both moments.

4. **Requirements software support.** The dedicated tool doing versioning and linking.

5. **Change control.** The formal request process for changing a baselined requirement.

6. Configuration management. And note it fired BECAUSE a change was accepted, the two elements chain together.

8. Configuration management again. The latest-version-clearly-identified duty.

Activity 15: Run the Change (Topic 6, after Slide 100)

Answer Key, Step by Step

Step 0, the coffee machine moment itself: the correct response to "no need for the forms" is a polite no. After baseline, EVERY change goes through the process, and "it is tiny" is exactly how scope control dies. Credit any table that opens with this; raise it yourself in debrief if nobody does, it is the most transferable lesson in the activity.

Step 1, Document: change to REQ-012, extend limit from three days to seven; requested by the Operations Manager, with date; reason: customer complaints about the three-day limit. What, who, when, why, on the form. The audit trail starts here, and exists even if the change is later rejected.

Step 2, Analyse: REQ-019 must be checked, price recalculation now spans up to seven days, do the pricing rules still work? REQ-024 becomes MORE likely to bite, a seven-day extension collides with next bookings far more often than three, what happens on collision? The booking screen changes; two test scripts change; and there may be a business exposure question: seven days of extension means vehicles unavailable longer, does fleet capacity cope in summer? Point out in debrief that the cross-references made this analysis fast, Activity 14's element two earning its keep.

Step 3, Consult: the Operations Manager as requester and likely owner; the business owner of bookings (the REQ-024 collision rule is theirs); developers or architect for build effort on screen and pricing; testers for the script updates; arguably the sponsor if fleet capacity is a commercial issue. Any three with a sensible "what they would say" earns it.

Step 4, Decide (recommend): model answer: "Recommend approval subject to the REQ-024 collision rule being confirmed with the bookings owner and the two test scripts being updated; estimated impact minor." Equally creditable: a compromise (five days), or approval deferred pending fleet capacity data. The mark is a recommendation grounded in the step two analysis, plus the explicit understanding that the change AUTHORITY decides, not the BA and not the requester.

Activity 16: Vertical or Horizontal? (Topic 6, after Slide 98)

The Six Traces and Answers

1. REQ-021 (check availability online) traces up to the objective: cut branch phone volume by 30%. VERTICAL, requirement to business objective.

2. REQ-017 traces back to the Workshop Supervisor's workshop of 5 June, and forward to test script TS-09 and the delivered inspection screen. HORIZONTAL, origin through to delivery and back.

3. Asked "why does this requirement exist at all?", the BA points to the business objective it serves. VERTICAL.

4. Asked "was this requirement actually delivered and tested?", the BA follows it forward to its test results. HORIZONTAL.

5. A regulation changes, and the team finds every affected requirement by following the links down from the legal parent requirement. VERTICAL, the hierarchy slide earning its keep.

6. An auditor asks to see the journey of REQ-012 from the person who raised it to the screen that implements it. HORIZONTAL, and it is only answerable because origin was recorded on day one, element three feeding traceability, everything in this topic connects.